Affine elimination: unreviewed proof packet

UNREVIEWED: two distinct regularity statements

General retained-row affine elimination

Graded quotient surjection

Conditional non-increase of top-part dreg

Translated k=2 control with field rows

Proposed exact count: 2p to p+1

Interval/rationality auxiliaries excluded

Independent proof pending. No strict full-chart or solving-degree inference.

Keep nonzero rows; preserve field premises and reconstruct eliminated variables.

Affine elimination and the regularity control

UNREVIEWED producer drafts. This report preserves both original derivations verbatim below. The general statement is a non-increase bound for the highest-homogeneous-part quotient, conditional on finite dimension and retention of every nonzero substituted row. The strict two-summand control excludes interval and rationality auxiliaries. Neither statement supplies a full-chart strict bound or a solving-degree comparison.

The original 106-cell coordinate and six-cell coefficient comparisons remain unchanged. Complete projected outputs, reconstruction, failed work and certificate checking remain required for any later SAT comparison. No suite, CAS, solver, census or scientific run was executed.

Affine elimination and top-part regularity

Root derivation, 2026-11-01. UNREVIEWED; no suite, solver or measurement. This is a general algebraic lemma and an implementation constraint for the full-chart reduction, not a claim of a strict regularity drop on every chart.

Metric. Use the degree of regularity defined by the vanishing threshold of the graded quotient by highest homogeneous input parts (Caminata and Gorla, arXiv:2112.05579v2, Definition 1.4). This differs from the solving degree of a Macaulay algorithm. The following proof is derived here; the cited paper supplies the metric definition, not this chart application.

Setup. Let K be a field, R=K[x,z1,...,zn], and S=K[z1,...,zn]. Let ell(z) be homogeneous linear, c a scalar, and L=x-ell(z)-c. For F={L,f1,...,fr}, form G={gi=fi(ell(z)+c,z)}, omitting only identically zero polynomials. Retain every other gi, including algebraically redundant rows. Let I be the ideal generated by the highest homogeneous parts of F and J the analogous ideal for G.

Claim. There is a graded surjection R/I onto S/J. Thus the Hilbert function of S/J is at most the Hilbert function of R/I in every degree. If R/I has finite dimension, dreg(G)<=dreg(F). This claim is non-strict; it neither compares running time nor states equality of solving degrees.

Proof. Put H=x-ell(z), and let psi:R->S be the graded substitution x->ell(z). Its kernel is (H), contained in I because H is the highest homogeneous part of L. For every fi of degree d, the degree-d component of gi is psi(fi_top): constants in the affine substitution affect only lower degrees. If gi still has degree d, this component is gi_top and belongs to J. If its degree drops, or gi vanishes, the component is zero and also belongs to J. Consequently psi(I) is contained in J. Since H belongs to I, R/I is isomorphic as a graded ring to S/psi(I), which surjects onto S/J. The dimension inequality and vanishing-threshold bound follow. A nonzero constant gi yields J=S and is covered too. The case with no z variables uses the same quotient convention.

Solution correspondence. Substitution and reconstruction x=ell(z)+c are inverse maps on solution sets over K and its extensions. This is separate from the graded-quotient argument. Over F_p, explicit field equations remain part of F and G. If all coefficients of ell and c lie in F_p, the substituted x field equation equals sum_i ell_i*(z_i^p-z_i). It may be dropped when all remaining z field equations are kept: its highest part is already in their highest-part ideal. For any other omitted row, require its top part to lie in the retained top-part ideal; ordinary ideal membership alone is insufficient.

Deletion counterexample. In K[z], the systems {z^2+z,z^2,z} and {z^2+z,z^2} define the same ideal (z), because z is the difference of the first two rows. Their top-part ideals are respectively (z) and (z^2), giving dreg1 and dreg2. Therefore a generally redundant equation can carry exactly the low-degree information the metric detects.

Finite-target application. The chart sign equation A(xT)-yT*B(xT)=0 has coefficient1 on A0. Solve A0=yT*B(xT)-sum_{i>=1} Ai*xT^i, an affine expression in the remaining chart coefficients. Apply the substitution to EVERY norm, membership, rationality, stratum and explicit field equation. Preserve all nonzero substituted rows in the first arm. This removes one field variable, preserves solutions and does not raise total input degree. The lemma applies to top-part regularity of this exact representation, including full-chart constraints, provided the starting top-part quotient is finite-dimensional (explicit field equations suffice). It does not promise a strict drop for that full chart.

Relation to the earlier equation-saving draft. The identity E0=K*L-sum_{j>=1} xT^j*Ej permits removal of E0 from the ordinary ideal, but does not by itself prove that removing its substituted row preserves the top-part invariant. Keep it in the regularity-preserving arm unless a separate highest-part membership argument is supplied. The translated k2 control has E0 vanish after substitution, so its earlier dreg2p to p+1 calculation is consistent with this rule; that control still awaits independent review.

Next check. Independently challenge the graded-surjection proof, the deletion counterexample, the coefficient1 target-sign pivot, and the retained-row contract. For future measurements, compare raw, affine-substituted-with-all-rows, and any separately justified row-pruned representation. Report solving degree, matrix dimensions and reconstruction work separately from this invariant.

A control separating top-part regularity from solving work

Root derivation, 2026-11-01. UNREVIEWED, UNEXECUTED. This is a proposed metric control for the full chart comparison, not a replacement for its complete k=0..5 cohort or a measured solver improvement.

Metric source. Caminata and Gorla, Solving degree, last fall degree, and related invariants, arXiv:2112.05579v2, Definition 1.4, define degree of regularity from when the ideal of highest homogeneous parts fills an entire graded component. Their Definition 1.1 defines solving degree using the Macaulay computation; these are distinct quantities. Source read on 2026-11-01: https://arxiv.org/pdf/2112.05579 . The construction and counts below are our derivation, not a claim attributed to that paper.

Public premises. Let p>3 be prime, E:y^2=x^3+alpha*x+beta nonsingular, and T=(a,b) a rational affine target. Put Z=X-a, so f(Z+a)=Z^3+c2*Z^2+c1*Z+b^2, with c2=3a and c1=3a^2+alpha. This is a polynomial-basis translation, not an assertion that the current short-Weierstrass source API accepts a nonzero quadratic curve coefficient. Use the k=2 finite-target chart A=A1*Z+A0, B=1, and U=Z^2+u1*Z+u0. Its norm is f(Z+a)-A^2=Z*U, and its target-sign equation is L=A0-b. Retained equations before substitution:

E2 = c2-A1^2-u1
E1 = c1-2*A1*A0-u0
E0 = b^2-A0^2
L = A0-b

Add the four explicit field equations v^p-v for v=A0,A1,u0,u1. No interval-membership equations are included in this diagnostic. The ideal of highest homogeneous parts is exactly (A0,A1^2,u0^p,u1^p). Indeed L kills A0, making E0 and the leading part of E1 redundant; A1^p is redundant modulo A1^2. The quotient's nonzero monomials have A1 exponent at most1 and u0,u1 exponents at most p-1. Its largest nonzero degree is 2p-1, so this definition gives dreg=2p. After substituting A0=b, omit the zero equation E0 and L. E2 remains quadratic, but E1 becomes c1-2b*A1-u0, a linear equation (also when b=0). Retain all three remaining field equations. The new top-part ideal is (A1^2,u0+2b*A1,u1^p); the equation u0^p is redundant after the linear substitution. Its largest nonzero quotient degree is p, giving dreg=p+1. This is an exact formal count under the specified field-equation representation. It has not been computer-checked or independently reviewed.

The source and reduced solutions correspond bijectively by A0=b. The redundant equation satisfies E0=-(A0+b)*L. Thus the observed change in this invariant is caused by exposing a lower-degree consequence through linear substitution. It does not establish a change in solving degree: an elimination solver can itself derive the same linear E1 consequence. No timing, matrix-rank computation, complete SAT model comparison or degree-of-regularity value for the full interval-constrained system is asserted.

Use as a control. A future metric implementation should recover the two displayed Hilbert staircases, then measure solving degree, matrix size and charged output recovery separately. Include b=0 and b!=0. Adding interval or rationality equations changes the top-part ideal and requires a fresh derivation; do not carry 2p or p+1 into the complete chart by assumption. The original 24-group correction suite and its frozen source remain unchanged.